

CDC 2026

```
library(dplyr)
library(tidyverse)
library(broom)
library(knitr)

complaints <- readRDS("complaints.rds")
census_zcta_2024 <- readRDS("census_zcta_2024.rds")

api_key <- "1587e10f3f8368c0fe2579d7bae54979c1ba79a4"

url <- paste0(
  "https://api.census.gov/data/2024/acs/acs5?",
  "get=NAME,",
  "B01003_001E,",
  "B19013_001E,",
  "B17001_001E,",
  "B17001_002E,",
  "B15003_001E,",
  "B15003_022E,",
  "B15003_023E,",
  "B15003_024E,",
  "B15003_025E,",
  "B23025_003E,",
  "B23025_005E,",
  "B02001_001E,",
  "B02001_002E,",
  "B02001_003E,",
  "B02001_005E,",
  "B02001_007E,",
  "B02001_008E,",
  "B03002_001E,",
  "B03002_012E,",
  "B01002_001E",
  "&for=zip%20code%20tabulation%20area:*",
  "&key=", api_key
)

census_raw <- read.csv(
  url,
  stringsAsFactors = FALSE,
  na.strings = c("", "null", "-6666666666", "-8888888888")
)
```

Cleaning and Prepping Data

```
complaints_2025 <- complaints %>%
  filter(
    `Date received` >= as.Date("2025-01-01"),
    `Date received` <= as.Date("2025-12-31"),
    Product == "Credit reporting or other personal consumer reports",
    `ZIP code` != "XXXXX"
  ) %>%
  select(`Date received`, Product, `ZIP code`)

head(complaints_2025, 10)
```

```
# A tibble: 10 × 3
  `Date received` Product                `ZIP code`
  <date>          <chr>                <chr>
1 2025-01-10      Credit reporting or other personal consumer repor... 32828
2 2025-01-09      Credit reporting or other personal consumer repor... 32828
3 2025-06-03      Credit reporting or other personal consumer repor... 32828
4 2025-06-03      Credit reporting or other personal consumer repor... 32828
5 2025-06-24      Credit reporting or other personal consumer repor... 75115
6 2025-06-25      Credit reporting or other personal consumer repor... 32828
7 2025-09-05      Credit reporting or other personal consumer repor... 19136
8 2025-09-05      Credit reporting or other personal consumer repor... 19136
9 2025-09-05      Credit reporting or other personal consumer repor... 19136
10 2025-09-20      Credit reporting or other personal consumer repor... 19136
```

Adding In Census Data

```
# Clean the ZCTA values
zcta_clean <- gsub("[^0-9]", "", as.character(census_raw$zip.code.tabulation.area))

# Make every ZCTA exactly 5 digits
zcta_clean <- sprintf("%05s", zcta_clean)
```

```
census_zcta_2024 <- data.frame(

  ZCTA = zcta_clean,

  population = as.numeric(census_raw$B01003_001E),

  median_household_income = as.numeric(census_raw$B19013_001E),

  poverty_percent =
    100 *
    as.numeric(census_raw$B17001_002E) /
    as.numeric(census_raw$B17001_001E),

  bachelors_or_higher_percent =
    100 *
```

```

(
  as.numeric(census_raw$B15003_022E) +
  as.numeric(census_raw$B15003_023E) +
  as.numeric(census_raw$B15003_024E) +
  as.numeric(census_raw$B15003_025E)
) /
as.numeric(census_raw$B15003_001E),

unemployment_percent =
  100 *
  as.numeric(census_raw$B23025_005E) /
  as.numeric(census_raw$B23025_003E),

white_percent =
  100 *
  as.numeric(census_raw$B02001_002E) /
  as.numeric(census_raw$B02001_001E),

black_percent =
  100 *
  as.numeric(census_raw$B02001_003E) /
  as.numeric(census_raw$B02001_001E),

asian_percent =
  100 *
  as.numeric(census_raw$B02001_005E) /
  as.numeric(census_raw$B02001_001E),

hispanic_percent =
  100 *
  as.numeric(census_raw$B03002_012E) /
  as.numeric(census_raw$B03002_001E),

other_race_percent =
  100 * as.numeric(census_raw$B02001_007E) /
  as.numeric(census_raw$B02001_001E),

two_or_more_races_percent =
  100 * as.numeric(census_raw$B02001_008E) /
  as.numeric(census_raw$B02001_001E),

median_age = as.numeric(census_raw$B01002_001E)
)

head(census_zcta_2024, 10)

```

	ZCTA	population	median_household_income	poverty_percent
1	00601	16669	19454	59.90141
2	00602	37233	21420	46.95903
3	00603	48448	20933	44.42137

4	00606	5163	20992	54.48383
5	00610	25357	24496	42.26457
6	00611	1383	20360	39.84093
7	00612	62908	24488	39.69674
8	00616	9717	21846	43.23851
9	00617	22584	25013	41.58499
10	00622	7664	26585	44.09563
	bachelors_or_higher_percent	unemployment_percent	white_percent	black_percent
1		18.19199	20.547945	79.806827
2		23.29432	5.133698	27.223162
3		28.39045	15.354331	72.884330
4		13.20944	0.000000	66.589192
5		22.73313	6.041044	18.716725
6		26.22222	18.721461	61.243673
7		29.93979	14.342392	13.259045
8		23.76396	18.200349	11.464444
9		25.05969	10.792526	8.204924
10		26.11718	12.006861	57.293841
	asian_percent	hispanic_percent	other_race_percent	two_or_more_races_percent
1	0.10798488	99.31010	8.362829	10.276561
2	0.11817474	97.23901	4.563156	66.220826
3	0.01857662	98.43337	12.169749	11.517503
4	0.23242301	99.67073	25.314739	7.728065
5	0.00000000	96.44280	9.303151	70.079268
6	0.00000000	97.39696	33.767173	2.458424
7	0.61518408	99.57398	24.766325	59.645832
8	1.13203664	99.54719	21.817433	62.529587
9	1.29295076	99.14098	22.343252	65.072618
10	0.26096033	98.40814	21.542276	15.396660
	median_age			
1	45.3			
2	46.6			
3	46.1			
4	47.9			
5	45.7			
6	55.9			
7	45.1			
8	49.0			
9	43.1			
10	48.1			

```
complaints_with_census <- complaints_2025 %>%
  mutate(
    ZCTA = sprintf("%05s", as.character(`ZIP code`))
  ) %>%
  left_join(
    census_zcta_2024,
    by = "ZCTA"
  )
```

```

complaints_by_zip <- complaints_2025 %>%
  mutate(
    ZCTA = sprintf("%05s", as.character(`ZIP code`))
  ) %>%
  count(ZCTA, name = "complaint_count") %>%
  left_join(
    census_zcta_2024,
    by = "ZCTA"
  ) %>%
  select(
    ZCTA,
    complaint_count,
    population,
    median_household_income,
    poverty_percent,
    bachelors_or_higher_percent,
    unemployment_percent,
    white_percent,
    black_percent,
    asian_percent,
    hispanic_percent,
    other_race_percent,
    two_or_more_races_percent,
    median_age
  ) %>%
  arrange(desc(complaint_count))

complaints_by_zip <- complaints_by_zip %>%
  mutate(
    complaints_per_1000 = (complaint_count / population) * 1000,
    complaint_level = case_when(
      complaints_per_1000 < 50 ~ 0,
      complaints_per_1000 >= 50 ~ 1
    )
  )

```

Linear Regression

```

linear_model <- lm(
  complaints_per_1000 ~
    median_household_income +
    poverty_percent +
    unemployment_percent +
    white_percent +
    black_percent +
    asian_percent +
    hispanic_percent +
    other_race_percent +

```

```

    two_or_more_races_percent +
    median_age,
    data = complaints_by_zip
  )

tidy(linear_model, conf.int = TRUE) |>
  knitr::kable(digits = 3)

```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	12.370	3.078	4.019	0.000	6.338	18.403
median_household_income	0.000	0.000	-5.112	0.000	0.000	0.000
poverty_percent	-0.355	0.021	-17.052	0.000	-0.396	-0.315
unemployment_percent	-0.091	0.039	-2.334	0.020	-0.168	-0.015
white_percent	-0.091	0.030	-3.054	0.002	-0.149	-0.032
black_percent	0.575	0.030	19.400	0.000	0.516	0.633
asian_percent	0.015	0.035	0.415	0.678	-0.054	0.084
hispanic_percent	0.202	0.021	9.519	0.000	0.160	0.243
other_race_percent	-0.162	0.044	-3.652	0.000	-0.250	-0.075
two_or_more_races_percent	0.005	0.043	0.119	0.905	-0.079	0.089
median_age	0.079	0.019	4.225	0.000	0.042	0.115

Testing for Multicollinearity

```

library(car)

vif(linear_model)

```

```

median_household_income      poverty_percent      unemployment_percent
      2.233776              2.421063              1.352320
white_percent                 black_percent          asian_percent
      34.937110             19.395418             4.820606
hispanic_percent             other_race_percent two_or_more_races_percent
      11.966282              9.557620              11.831709
median_age
      1.187580

```

```

complaints_by_zip <- complaints_by_zip %>%
  mutate(
    income_10k = median_household_income / 10000
  )

linear_model_no_race <- lm(

```

```

complaints_per_1000 ~
  income_10k +
  poverty_percent +
  unemployment_percent +
  median_age,
data = complaints_by_zip
)

tidy(linear_model_no_race, conf.int = TRUE) |>
  knitr::kable(digits = 3)

```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	19.132	1.200	15.944	0	16.780	21.484
income_10k	-0.407	0.054	-7.560	0	-0.513	-0.302
poverty_percent	0.103	0.024	4.327	0	0.056	0.149
unemployment_percent	0.567	0.045	12.473	0	0.478	0.656
median_age	-0.204	0.021	-9.636	0	-0.245	-0.162

```

model_black <- lm(
  complaints_per_1000 ~
    median_household_income +
    poverty_percent +
    unemployment_percent +
    median_age +
    black_percent,
data = complaints_by_zip
)

model_white <- lm(
  complaints_per_1000 ~
    median_household_income +
    poverty_percent +
    unemployment_percent +
    median_age +
    white_percent,
data = complaints_by_zip
)

model_asian <- lm(
  complaints_per_1000 ~
    median_household_income +
    poverty_percent +
    unemployment_percent +
    median_age +
    asian_percent,
data = complaints_by_zip
)

```

```

model_hispanic <- lm(
  complaints_per_1000 ~
    median_household_income +
    poverty_percent +
    unemployment_percent +
    median_age +
    hispanic_percent,
  data = complaints_by_zip
)

```

```

model_other_race <- lm(
  complaints_per_1000 ~
    median_household_income +
    poverty_percent +
    unemployment_percent +
    median_age +
    other_race_percent,
  data = complaints_by_zip
)

```

```

model_two_or_more <- lm(
  complaints_per_1000 ~
    median_household_income +
    poverty_percent +
    unemployment_percent +
    median_age +
    two_or_more_races_percent,
  data = complaints_by_zip
)

```

```

bind_rows(
  tidy(model_white, conf.int = TRUE) %>%
    filter(term == "white_percent") %>%
    mutate(race = "White"),

  tidy(model_black, conf.int = TRUE) %>%
    filter(term == "black_percent") %>%
    mutate(race = "Black"),

  tidy(model_asian, conf.int = TRUE) %>%
    filter(term == "asian_percent") %>%
    mutate(race = "Asian"),

  tidy(model_hispanic, conf.int = TRUE) %>%
    filter(term == "hispanic_percent") %>%
    mutate(race = "Hispanic"),

  tidy(model_other_race, conf.int = TRUE) %>%
    filter(term == "other_race_percent") %>%

```

```

mutate(race = "Other race"),

tidy(model_two_or_more, conf.int = TRUE) %>%
  filter(term == "two_or_more_races_percent") %>%
  mutate(race = "Two or more races")
) %>%
select(race, estimate, std.error, statistic, p.value, conf.low, conf.high) %>%
knitr::kable(digits = 3)

```

race	estimate	std.error	statistic	p.value	conf.low	conf.high
White	-0.412	0.006	-63.469	0	-0.425	-0.400
Black	0.586	0.008	75.059	0	0.571	0.602
Asian	0.110	0.022	5.019	0	0.067	0.152
Hispanic	0.053	0.008	6.781	0	0.038	0.069
Other race	0.100	0.018	5.470	0	0.064	0.136
Two or more races	0.089	0.016	5.689	0	0.058	0.120